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Question Bank

23INMCA308 - Design & Analysis of Algorithms

Course Type	Course Nature	CA Conduct	System	L	T	P	Credits	CA Total	CA Pass	SEE Total	SEE Pass	Total Pass
Theory	1	End Semester	Mark	3	1	0	4	40	24	60	24	50

Question Bank Summary

Sect. Part A	Sect. Part B	Easy	Med.	Chall.	Th.	Appli.
36	36	19	37	16	60	12

Part A

#	Unit	Question	COS	Categorized
1	1.1	Explain about Time and Space Complexity	CO1	Easy - Understanding - T
2	1.1	Discuss about Asymptotic Notations	CO1	Easy - Understanding - T
3	1.1	Explain about Recurrence Equations. Give one Recurrence Equation and explain it	CO1	Easy - Understanding - T
4	1.1	Given an algorithm with a time complexity of $O(n \log n)$ and a space complexity of $O(n)$, apply these complexities to a scenario where $n=1000$. Calculate the approximate number of operations and the memory used.	CO1	Challenging - Applying - A
5	1.1	Explain the methods of solving Recurrence Equations	CO1	Medium - Remembering - T
6	1.1	Explain the Masters method for solving recurrence equations?	CO1	Easy - Understanding - T
7	1.1	Explain the iteration method for solving recurrence equations ?	CO1	Easy - Understanding - T
8	1.1	Apply the iteration method to solve a given recurrence equation	CO1	Medium - Applying - A
9	1.1	Demonstrate how the recursion tree method can be used to solve a recurrence equation	CO1	Easy - Understanding - T
10	1.1	Two programs perform the same task, but one is tested on a computer while the other is analyzed using mathematical expressions. Identify and explain Priori and Posteriori analysis of algorithms based on this scenario.	CO1	Medium - Applying - T
11	1.1	A mobile application must run efficiently on devices with limited memory. Explain the importance of space complexity in selecting an algorithm for this application.	CO1	Medium - Applying - T

12	1.1	An algorithm takes $5n^2+3n+10$ steps for an input of size n . Estimate the time complexity using asymptotic notation and justify your answer.	CO1	Medium - Understanding - T
13	2.1	A software application processes a large dataset by breaking it into smaller parts, solving each part independently, and then combining the results. a) Identify the algorithmic strategy used. b) Mention one advantage of this strategy.	CO2	Easy - Understanding - T
14	2.1	A student is searching for a roll number in a sorted list of 1,000 students using a fast searching technique. Why is Binary Search more efficient than Linear Search in this scenario?	CO2	Medium - Applying - T
15	2.1	A file system needs to sort very large datasets stored on disk efficiently. Which sorting algorithm is suitable here and why is Merge Sort preferred?	CO2	Medium - Applying - T
16	2.1	An application requires fast average-case sorting of data stored in memory. Name the sorting technique used and state the role of the pivot element.	CO2	Easy - Understanding - T
17	2.1	A temperature monitoring system needs to find both the highest and lowest readings efficiently from a dataset. How does the Divide and Conquer approach reduce the number of comparisons?	CO2	Easy - Understanding - T
18	2.1	A developer designs a generic template for algorithms that make the locally optimal choice at each step. Which strategy does this represent, and what is its main idea?	CO2	Easy - Understanding - T
19	2.1	A traveler has a bag with limited capacity and wants to maximize the value of items carried. Identify the algorithmic problem and mention one real-life application.	CO2	Medium - Applying - A
20	2.1	A city planning department wants to connect all locations with minimum total road cost. Which algorithm should be used and why is it suitable?	CO2	Medium - Understanding - T
21	2.1	A network engineer wants to connect computers with the least total cable cost, considering edges in increasing order of weight. Which algorithm follows this approach?	CO2	Easy - Remembering - T
22	2.1	Given a dense graph, which algorithm is more suitable: Prim's or Kruskal's? Justify your answer briefly.	CO2	Challenging - Analysing - A
23	2.1	A company wants to schedule jobs with deadlines to maximize profit using a greedy approach. What is the objective of the Job Sequencing Problem?	CO2	Medium - Understanding - T
24	2.1	A programmer wants to design a generic structure for algorithms that divide a problem, solve subproblems, and combine solutions. What is control abstraction, and why is it useful in Divide and Conquer algorithms?	CO2	Easy - Understanding - T
25	3.1	A software engineer is learning algorithm design techniques. Define Dynamic Programming and state one situation where it is preferred over other techniques.	CO3	Easy - Remembering - T
26	3.1	A student is asked to optimize a recursive solution that recalculates the same subproblems. Explain why Dynamic Programming is suitable for such problems.	CO3	Easy - Understanding - T
27	3.1	While developing a program, a developer notices that a problem can be divided into smaller independent parts. Explain the Divide and Conquer technique with a suitable example.	CO3	Medium - Understanding - T
28	3.1	A company is choosing between Divide and Conquer and Dynamic Programming for solving a problem with overlapping subproblems. Which technique would you suggest and why?	CO3	Medium - Applying - T
29	3.1	An algorithm repeatedly solves the same subproblems using recursion. Analyze how Dynamic Programming improves efficiency compared to Divide and Conquer in this scenario.	CO3	Challenging - Analysing - T
30	3.1	A researcher wants to ensure that solving subproblems leads to an optimal overall solution. State and explain the Principle of Optimality.	CO3	Easy - Understanding - T
31	3.1	A navigation system needs to find the shortest path between every pair of cities. Name the algorithm used and mention its main objective.	CO3	Challenging - Applying - T

32	3.1	In a road network application, all-pair distance information is required. Explain the All-Pair Shortest Path problem in brief.	CO3	Easy - Understanding - T
33	3.1	A salesman must visit all cities exactly once and return to the starting city with minimum travel cost. Identify the problem and mention the technique commonly used to solve it.	CO3	Medium - Applying - A
34	3.1	Two algorithms are given: one uses Divide and Conquer and the other uses Dynamic Programming. Differentiate between Divide and Conquer and Dynamic Programming based on subproblem overlap.	CO3	Challenging - Analysing - T
35	3.1	A student observes that Dynamic Programming solutions are often stored in tables. Explain the purpose of storing intermediate results in Dynamic Programming.	CO3	Easy - Understanding - T
36	3.1	A logistics company wants an efficient solution for route planning with minimum cost. Explain why Dynamic Programming is suitable for solving the Traveling Salesperson Problem.	CO3	Medium - Applying - T

Part B

#	Unit	Question	COS	Categorized
1	1.1	A student is asked to design a step-by-step solution to sort a list of numbers before coding. a) Define an algorithm. b) Explain any four properties of a good algorithm with reference to the above scenario.	CO1	Medium - Understanding - T
2	1.1	A software engineer explains that an algorithm must always produce correct results and terminate after a finite number of steps. Identify and explain the essential properties of algorithms that justify this statement.	CO1	Medium - Understanding - T
3	1.1	A school management system needs to search for a student record in a database. Based on this scenario: a) Identify suitable types of algorithms that can be used. b) Explain how the choice of algorithm affects performance.	CO1	Medium - Applying - T
4	1.1	A programmer wants to estimate the performance of an algorithm before implementation and also measure its actual execution time after implementation. Explain Priori and Posteriori analysis of algorithms with examples.	CO1	Medium - Applying - T
5	1.1	While designing a mobile application, memory usage becomes a concern. a) Define space complexity. b) Explain the factors affecting space complexity of an algorithm.	CO1	Medium - Understanding - T
6	1.1	An algorithm contains a single loop that executes n times and performs a constant-time operation inside the loop. Estimate the time complexity of the algorithm and justify your answer.	CO1	Medium - Applying - T
7	1.1	for i = 1 to n for j = 1 to n print(i, j) For the above code snippet, analyze and estimate the time complexity of the algorithm using step-count method.	CO1	Challenging - Analysing - A
8	1.1	A recursive algorithm is represented by the recurrence relation: $T(n)=T(n-1)+n$ Solve the recurrence relation using the Iterative Method and determine its time complexity.	CO1	Challenging - Analysing - A
9	1.1	A divide-and-conquer algorithm is expressed as: $T(n)=2T(n/2)+n$ Solve the recurrence using the Recurrence Tree Method and find the order of growth.	CO1	Challenging - Analysing - A
10	1.1	A student is confused about selecting the appropriate asymptotic notation to express algorithm efficiency. Explain Big-O, Big-Ω, and Big-Θ notations with suitable examples.	CO1	Challenging - Applying - A

11	1.1	Consider the recurrence relation of an algorithm: $T(n)=3T(n/2)+n$ Apply the Master Theorem to determine the time complexity of the algorithm.	CO1	Medium - Analysing - A
12	1.1	Use the recursion tree method to solve the recurrence relation $T(n)=3T(n/3)+n$. Provide the detailed steps and final solution.	CO1	Medium - Applying - A
13	2.1	A software company needs to process a very large dataset by breaking it into smaller independent tasks and combining the results efficiently. a) Identify the algorithmic strategy suitable for this scenario. b) Explain how the Divide and Conquer technique works with a suitable example.	CO2	Medium - Understanding - T
14	2.1	While designing an algorithm, a programmer wants a generic structure that can be reused for different Divide and Conquer problems. Explain the control abstraction for Divide and Conquer and describe how it improves algorithm design.	CO2	Medium - Understanding - T
15	2.1	A library system stores book IDs in sorted order. To quickly locate a book, an efficient searching technique is required. a) Identify the suitable algorithm. b) Explain its working and time complexity.	CO2	Medium - Understanding - T
16	2.1	An online examination system needs to sort student scores efficiently even when the number of students is very large. Explain how Merge Sort solves this problem using the Divide and Conquer approach.	CO2	Challenging - Understanding - T
17	2.1	A database administrator prefers an in-place sorting algorithm with good average-case performance. Explain the working of Quick Sort and discuss its best and worst-case time complexity.	CO2	Medium - Understanding - T
18	2.1	In a sports analytics application, the highest and lowest scores from a large dataset must be found efficiently. Explain how the Divide and Conquer method can be used to find the maximum and minimum elements.	CO2	Medium - Applying - T
19	2.1	A developer wants to design algorithms that make optimal choices at every step without reconsidering previous decisions. Explain the control abstraction for the Greedy strategy and its characteristics.	CO2	Medium - Understanding - T
20	2.1	A delivery company has limited storage capacity and wants to maximize profit by selecting the most valuable packages. Explain how the Greedy approach can be applied to the Knapsack Problem.	CO2	Medium - Applying - T
21	2.1	A telecommunication company wants to connect multiple cities with minimum total cable cost. Explain how Prim's Algorithm can be used to find the minimum spanning tree for this problem.	CO2	Challenging - Applying - A
22	2.1	In a road construction project, engineers want to connect towns with minimum cost without forming cycles. Describe the working of Kruskal's Algorithm and explain how it avoids cycles.	CO2	Challenging - Applying - T
23	2.1	Given two different minimum spanning tree algorithms, analyze the situations where Prim's Algorithm is preferred over Kruskal's Algorithm.	CO2	Challenging - Analysing - T
24	2.1	A company has multiple jobs with deadlines and profits but can process only one job at a time. Explain how the Job Sequencing Problem is solved using the Greedy strategy to maximize profit.	CO2	Medium - Applying - T
25	3.1	A logistics company frequently solves optimization problems where subproblems overlap and results are reused. a) Identify the suitable algorithmic technique. b) Explain its basic idea with an example.	CO3	Medium - Understanding - T
26	3.1	A large dataset is recursively divided into smaller parts for efficient processing. Explain the Divide and Conquer strategy, its three main steps, and illustrate with a suitable algorithm.	CO3	Medium - Understanding - T
27	3.1	A software engineer is confused between using Divide and Conquer or Dynamic Programming for solving a problem with repeated subproblems. Compare the two techniques with examples and justify when Dynamic Programming is preferred.	CO3	Challenging - Analysing - T
28	3.1	A route planning application computes optimal paths by combining solutions of smaller paths. Define the Principle of Optimality and explain its role in Dynamic Programming with an example.	CO3	Medium - Understanding - T
29	3.1	An algorithm designer wants to implement a Dynamic Programming solution for an optimization problem. Explain the systematic steps involved in designing a Dynamic Programming algorithm.	CO3	Medium - Applying - T

30	3.1	A city traffic system needs the shortest paths between all pairs of junctions. Explain how the Floyd–Warshall algorithm solves the All Pair Shortest Path problem.	CO3	Easy - Understanding - T
31	3.1	An application requires fast computation but has limited memory resources. Discuss the time and space trade-offs in Dynamic Programming approaches.	CO3	Challenging - Analysing - T
32	3.1	A delivery service must visit multiple cities exactly once and return to the starting point at minimum cost. Explain the Travelling Sales Person Problem and describe how Dynamic Programming can be used to solve it.	CO3	Medium - Applying - T
33	3.1	A problem is repeatedly solving the same subproblems during execution. Explain the concept of overlapping subproblems and how Dynamic Programming handles it.	CO3	Medium - Understanding - T
34	3.1	An engineer notices that the optimal solution to a problem can be constructed from optimal solutions of its subproblems. Explain optimal substructure and its significance in Dynamic Programming.	CO3	Easy - Understanding - T
35	3.1	A navigation system must choose between multiple algorithms for route optimization. Analyze and justify whether Dynamic Programming or Divide and Conquer is more suitable.	CO3	Challenging - Evaluating - T
36	3.1	A developer finds Dynamic Programming efficient but resource-intensive. Explain the limitations of Dynamic Programming and situations where it may not be suitable.	CO2	Medium - Understanding - T